Noise and Outliers

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Teach arithmetic and data decisions with separate evidence.

Supply all records and scripts, distinguish the two replacement scenarios and reserve the fresh list for assessment.

Keep one habit: Require a sum and denominator for the mean, sorting for the median and provenance for a repair.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Calculate **mean** and **median**

Teach Week 17 as a two-session investigation using fictional data. Five fictional activity durations are 2, 3, 3, 4 and 18 minutes. Most are near 3, but one is much longer. Different summaries reveal different aspects.

Keep all values while investigating whether 18 is a genuine long activity or a recording error. Assess the three stated outcomes through the independent task, not by copying the worked result.

By the end, I can calculate **mean** and **median** and **investigate** an **unusual value** before changing it.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Python 3 editor with a fresh local practice file
- Paper trace table and the student workbook
- Fictional **input** records from this lesson; no real student identifiers

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Over**fitting** Intuition. Give an example and a mistake that the idea helps you avoid.

Look for: Compare **training** and validation **evidence** for model capacity

Reference example, not the only acceptable wording: Teacher explanation: Choose the line with validation MAE 3. The curve **training** MAE 0 does not compensate for validation MAE 8.

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a root question and leaf **prediction**s from **training** examples

Reference example, not the only acceptable wording: The root asks near, with branches 0 and 1. Leaves predict clear for 0 and blocked for 1. Depth is one.

Training labels determined both leaf majorities and the error comparison that selected near rather than tall.

Without opening the old lesson, explain one key idea from **Training** and Testing. Give an example and a mistake that the idea helps you avoid.

Look for: Choose a split unit that matches the intended **prediction** claim

Reference example, not the only acceptable wording: **Training** fits the model; validation compares development choices; test evaluates after choices are fixed.

Repeatedly selecting by the test score uses that set for development, so it is no longer an untouched final evaluation.

9.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Overfitting Intuition

Decision Trees

Training and Testing

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Page counts are 72,80,96,960. The last book may be an anthology. Should a cleaning script delete it automatically? Give a defensible investigation and document the outcome.

Concept to explain

Mistakes and **unusual value**s should be **investigate**d, not automatically erased.

Term	Meaning
noise	Variation or interference that obscures the information relevant to a task.

Term	Meaning
outlier	An observation far from a typical pattern under a stated comparison; it is not automatically an error.
duplicate	A repeated record or copy; copies are not independent new observations.
investigate	Collect and examine evidence to answer a question rather than assuming a cause.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach summaries as different questions about the same records

> Supply full scripts and distinguish a what-if from an authorised correction.

Introduce the five durations and odd-count script

Five fictional activity durations are 2, 3, 3, 4 and 18 minutes. Most are near 3, but one is much longer. Different summaries reveal different aspects.

Keep all values while investigating whether 18 is a genuine long activity or a recording error.

```
values = [2, 3, 3, 4, 18]
ordered = sorted(values)
mean = sum(values) / len(values)
median = ordered[len(ordered) // 2]
print(mean)
print(median)
```

Explain sorting, mean and the odd-count limitation

The script prints 6.0 and 3. sorted returns a sorted list without changing the original. This **median** expression works for a nonempty odd-length list only. For four sorted values, average the second and third.

Do not use this odd-length shortcut for every dataset.

Extend explicitly to even and empty cases

```
def summaries(values):
    if not values:
        return None, None
    ordered = sorted(values)
    middle = len(ordered) // 2
    if len(ordered) % 2 == 1:
        median = ordered[middle]
    else:
        median = (ordered[middle - 1] + ordered[middle]) / 2
    mean = sum(values) / len(values)
    return mean, median

for values in [[2, 3, 3, 4, 18], [8, 2, 4, 6], []]:
    print(summaries(values))
```

Collect the18→8 what-if before explaining

Replace 18 with 8. Calculate the new mean and median.

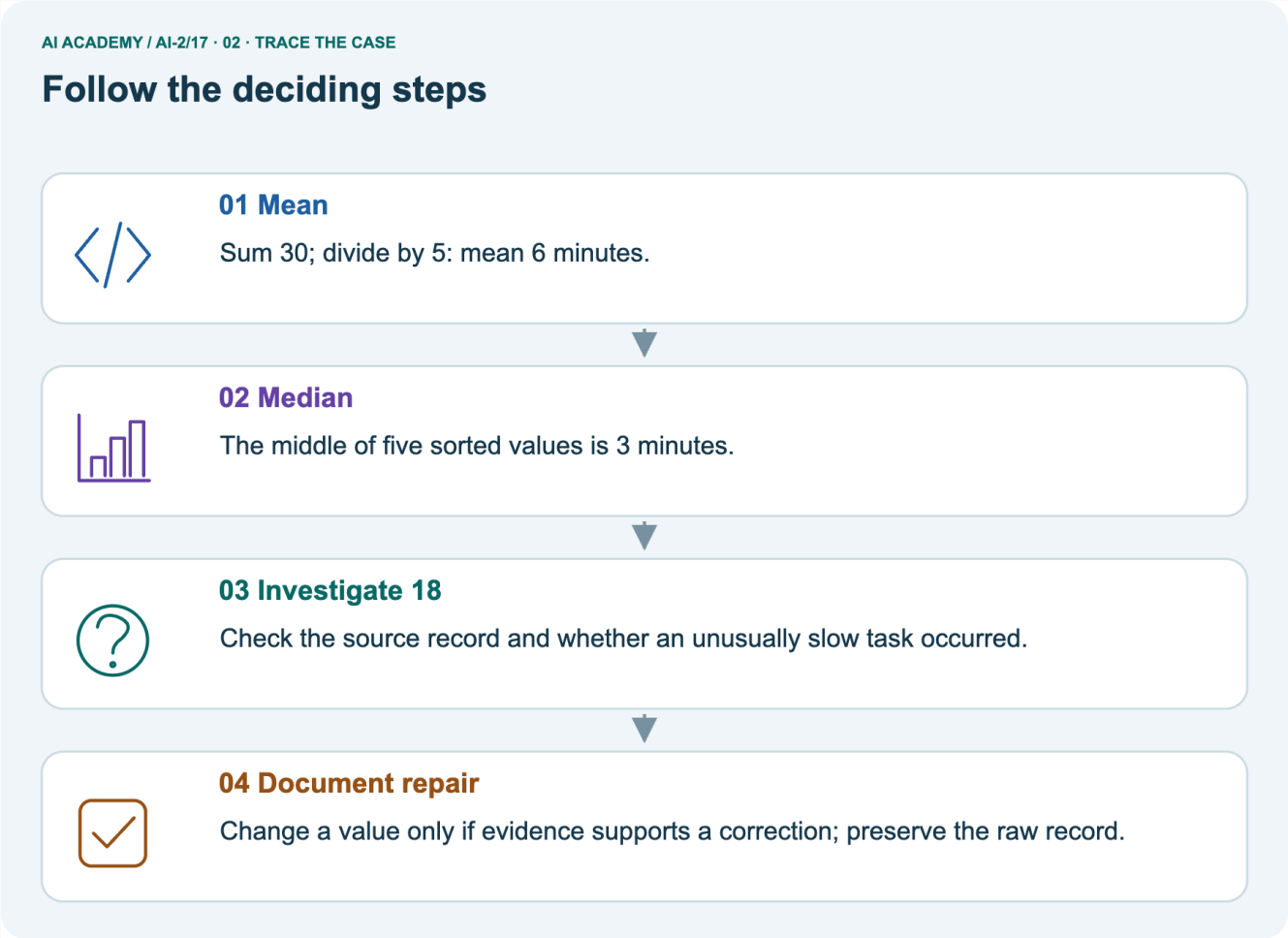
Read the pictured baseline and unconfirmed source

A fictional library records five book-**process**ing times in minutes.

Sorted times: 2,3,3,4,18.

All are recorded measurements; no **evidence** yet shows 18 is a mistake.

Trace the summaries and investigation decision



Follow the labelled picture with the matching explanation. Open the vector diagram

The **mean** uses every value and is pulled upward by 18. The **median** identifies the middle recorded time and stays at 3. Neither statistic declares the unusual **observation** invalid.

Deleting it simply because it changes the **mean** would hide potentially real slow work and change the question being answered.

Five task times do not estimate all library **process**ing times reliably or prove which summary suits every decision.

Supply the authorised18→3 repair

Change: An authorised log confirms 18 was a transcription error for 3.

Prediction: Corrected **mean** is 3 minutes and **median** is 3.

Reason: Values become 2,3,3,4,3 with sum 15; sorted middle remains 3.

Use the picture to elicit the evidence limit

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Most books: 40–300 pages; one record: 3,200 pages.

2. Check the source: extra zero, box set, or real long work?

3. Repair confirmed error; keep real rare case and document it.

4. Retest to learn whether the system handles legitimate extremes.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Recorded lengths are 90,95,100,105,900 pages.

Is 900 automatically wrong?

Exact explanation to model

No. It is an **outlier**, but may be a real large book. Check catalogue, units and entry **process**.

Explain the added worked example

Key terms

Term	Meaning
Mean	Sum of numeric values divided by their count
Median	The middle of sorted values or mean of the two middle values
Unusual value	A value far from most others in a dataset
Variation	Differences among recorded values

Model the reasoning aloud

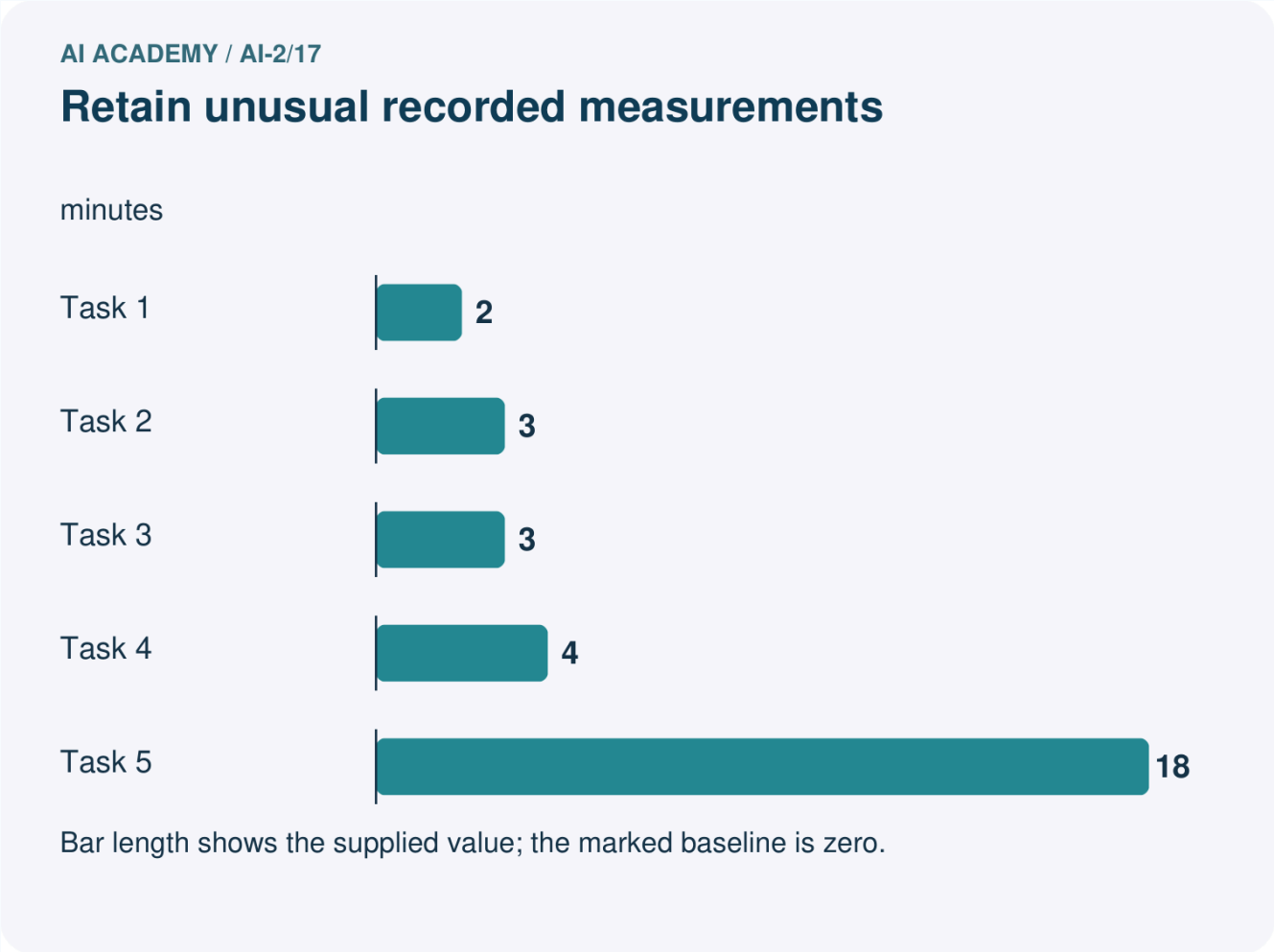
Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: Noise and Outliers

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/17

Noise and Outliers

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Mean

Sum 30; divide by 5: mean 6 minutes.



2. Median

The middle of five sorted values is 3 minutes.



3. Investigate 18

Check the source record and whether an unusually slow task occurred.



4. Document repair

Change a value only if evidence supports a correction; preserve the raw record.

CHANGE ONE THING

An authorised log confirms 18 was a transcription error for 3.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Mean	Sum 30; divide by 5: mean 6 minutes.
2. Median	The middle of five sorted values is 3 minutes.
3. Investigate 18	Check the source record and whether an unusually slow task occurred.
4. Document repair	Change a value only if evidence supports a correction; preserve the raw record.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

Prompt before revealing

Completion task: Find **median**s of [9, 1, 3] and [8, 2, 4, 6].

Expected reasoning and fading help

Teacher check: Sorted first list 1, 3, 9 has **median** 3. Sorted second 2, 4, 6, 8 has **median** (4 + 6)/2 = 5.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Check the misconception before discussion

Use the worked case for **Mean median** and **unusual value**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Removing 18 merely because it raises the **mean** would hide **evidence**. Check collection notes, keep an audit trail and compare summaries with and without it only with a clear explanation.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding **variation**.

Use the pictorial check

If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?

Median 3 describes the middle; also report the unusual 18-minute task and the **mean** 6 rather than concealing spread.

Changed-case reasoning: Corrected **mean** is 3 minutes and **median** is 3. Values become 2,3,3,4,3 with sum 15; sorted middle remains 3.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Demonstrate and guide practice

Board sequence

Write the given facts: Recorded lengths are 90,95,100,105,900 pages.

Write the question to explain: Is 900 automatically wrong?

Reveal after **prediction**s: No. It is an **outlier**, but may be a real large book. Check catalogue, units and entry **process**.

Use the earlier changed question as optional later practice. Reserve Compare summaries independently for the first independent attempt before hints or feedback.
</details>

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Sum 20; **mean** 4; **median** remains 3. Exit: No. Zero would be a measured summary; no numeric **observation**s are available.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

Baseline2,3,3,4,18 gives sum30,mean6,median3. Hypothetical18→8 gives sum20,mean4,median3. Authorised18→3 gives sum15,mean3,median3. W1 sum30/count5=6; W2 medians3 and5. General script outputs(6.0,3),(5.0,5.0),(None,None). Audit R1: correct18→3 with log; R2: retain verified25; R3: exclude proven duplicate from analysis while retaining raw source; R4: ask/review unavailable duration. These are separate audit events, not a combined statistical sample. Reserve the fresh five-value summary comparison for W4.

Run and record

1. Predict → Test → Record → Explain
2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.
3. Use the four supplied audit records. Choose keep, correct, exclude a confirmed duplicate from analysis, or ask for evidence; justify every action from the supplied source facts and preserve the raw record.
4. What did the hardest test teach you that the normal test did not?

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim

Record	What to capture
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

A successful run proves the conclusion: Removing 18 merely because it raises the **mean** would hide **evidence**. Check collection notes, keep an audit trail and compare summaries with and without it only with a clear explanation.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: 2, 4, 4, 5, 15 has sum 30, **mean** 6 and **median** 4. Replacing 15 with 5 gives sum 20, **mean** 4 and **median** 4.

A convenient shortcut preserves **mean**ing: Taking the middle item of an unsorted list can give the wrong **median**. For [18, 2, 4, 3, 3], the middle item is 4, but the sorted **median** is 3. Sort first.

For an empty list, there is no **mean** or **median** under this lesson's rules.

Choose support from the response

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows	Next teaching action
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Compare summaries independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected summary comparison

> Keep the answer until after the first independent attempt.

Selected W4 and checked key

Selected case: For 1, 2, 2, 3 and 12, calculate **mean** and **median**. Then replace 12 with 2 and recalculate. Explain the change.

Original sum 20, **mean** 4, **median** 2. Revised sum 10, **mean** 2, **median** 2. The large value changes **mean** more than **median** in this example.

Judge calculation, explanation and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 mean key

Find the **mean** of 4, 4, 6, 6 and 10.

Sum 30 divided by 5 gives 6.

Add all five **observation**s: 4 + 4 + 6 + 6 + 10 = 30. Divide by the number of **observation**s, five, to obtain a **mean** of 6. Keep the count beside the calculation.

Do not divide by the three distinct values; repeated **observation**s still count.

W2 median keys

Find **median**s of [9, 1, 3] and [8, 2, 4, 6].

Sorted first list 1, 3, 9 has **median** 3. Sorted second 2, 4, 6, 8 has **median** (4 + 6)/2 = 5.

W3 source-check key

A learner deletes 18 because it is above the **mean**. What **evidence** is needed before correcting or excluding it?

Check original measurement and collection rules. Being above the **mean** alone is not an error. Retain the raw record and document any justified correction.

W5 empty-data key

A dataset is empty. Is reporting a **mean** of 0 justified?

No. Zero would be a measured summary; no numeric **observation**s are available.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

No. It is an **outlier**, but may be a real large book. Check catalogue, units and entry **process**.

Additional transfer practice question

Page counts are 40, 42, 43 and 900. Should 900 be deleted immediately? Give two explanations to **investigate**.

Reasoned practice answer

900 may be a genuine long book or a recording error. Check the original record and task scope before changing it; unusual does not automatically **mean** wrong.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Every **outlier** is an error.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Check edition, catalogue and source entry. Retain 960 if verified; correct only if **evidence** establishes an error, recording original value, new value and reason.

Automatic deletion can remove a legitimate category and distort the intended collection.

Explanations for the additional workbook examples

W1 Compute a mean

W2 Compute two medians

W3 Repair an explanation

W4 Compare summaries independently

For 1, 2, 2, 3 and 12, calculate **mean** and **median**. Then replace 12 with 2 and recalculate. Explain the change.

Original sum 20, **mean** 4, **median** 2. Revised sum 10, **mean** 2, **median** 2. The large value changes **mean** more than **median** in this example.

W5 Report missing evidence

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Keep legitimate unusual books so the recommender can serve unusual readers.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Investigate unusual recommender features and show **evidence**-backed repairs instead of removing inconvenient records.

Support while preserving the reasoning goal

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Order number cards physically. Extension: write a **median** function with explicit empty, odd and even cases and test each.

Independent W4 **evidence**: Original sum 20, **mean** 4, **median** 2. Revised sum 10, **mean** 2, **median** 2. The large value changes **mean** more than **median** in this example. Record whether the learner needed no help, a hint or explanation.

Fresh retry: 2, 4, 4, 5, 15 has sum 30, **mean** 6 and **median** 4. Replacing 15 with 5 gives sum 20, **mean** 4 and **median** 4.

Plan the delayed check

Suggested return points in this level: ai-2/18, ai-2/20, ai-2/23. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	

Record	Teacher evidence
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 18: Missing Data. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week 18: Missing Data. Carry forward the difference between an unavailable measurement and zero, preserving source **evidence** and unresolved review status.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Transparent Book Recommender

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Calculate **mean** and **median**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

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> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

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>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice and assessment reminders

Before class, solve the supported practice: Find **median**s of [9, 1, 3] and [8, 2, 4, 6].

Reasoned practice answer: Sorted first list 1, 3, 9 has **median** 3. Sorted second 2, 4, 6, 8 has **median** (4 + 6)/2 = 5.

Check every learner before discussion: Use the worked case for **Mean median** and **unusual value**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Removing 18 merely because it raises the **mean** would hide **evidence**. Check collection notes, keep an audit trail and compare summaries with and without it only with a clear explanation.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: An authorised log confirms 18 was a transcription error for 3. Require a **prediction** and a reason before comparing. Expected result: Corrected **mean** is 3 minutes and **median** is 3.

Reason: Values become 2,3,3,4,3 with sum 15; sorted middle remains 3.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

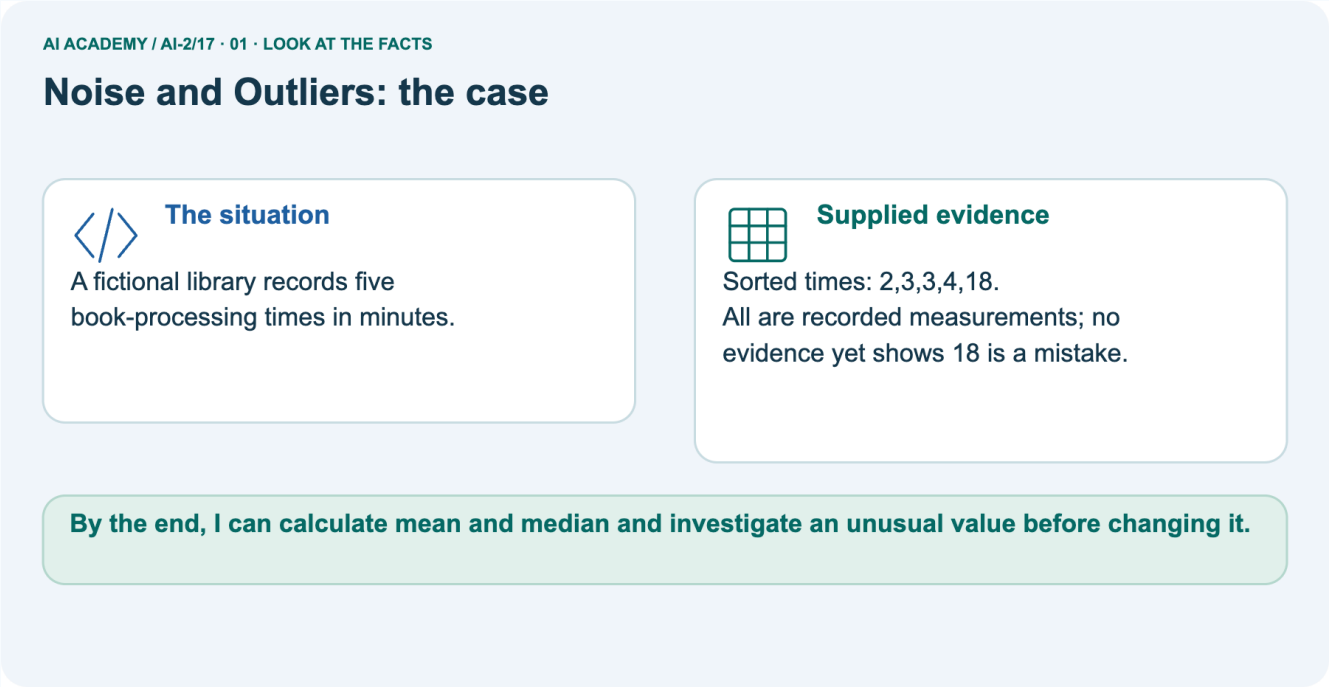
<details>
<summary>Reference: teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Noise and Outliers: the case



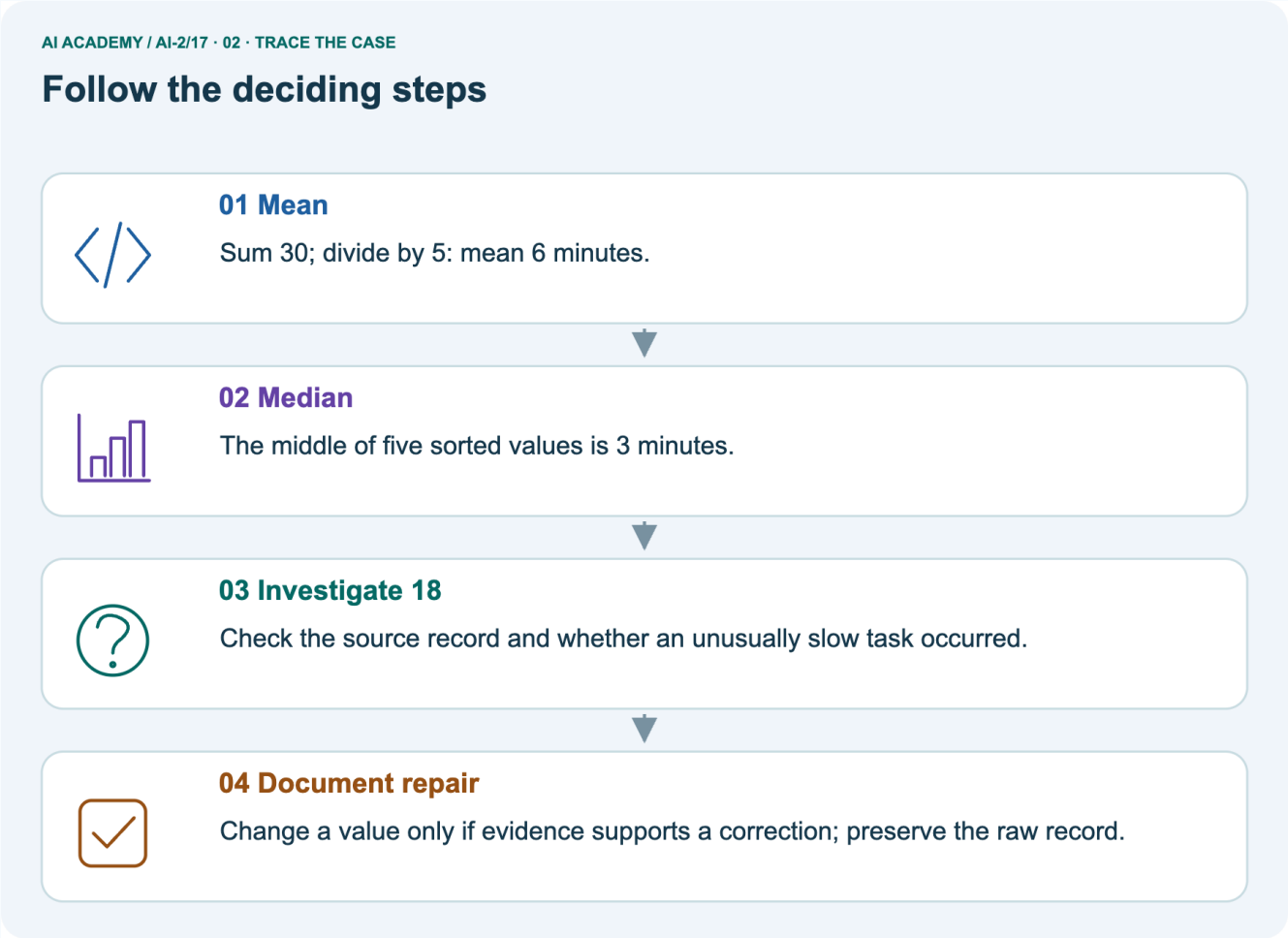
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional library records five book-**process**ing times in minutes. Supplied facts: Sorted times: 2,3,3,4,18. All are recorded measurements; no **evidence** yet shows 18 is a mistake.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



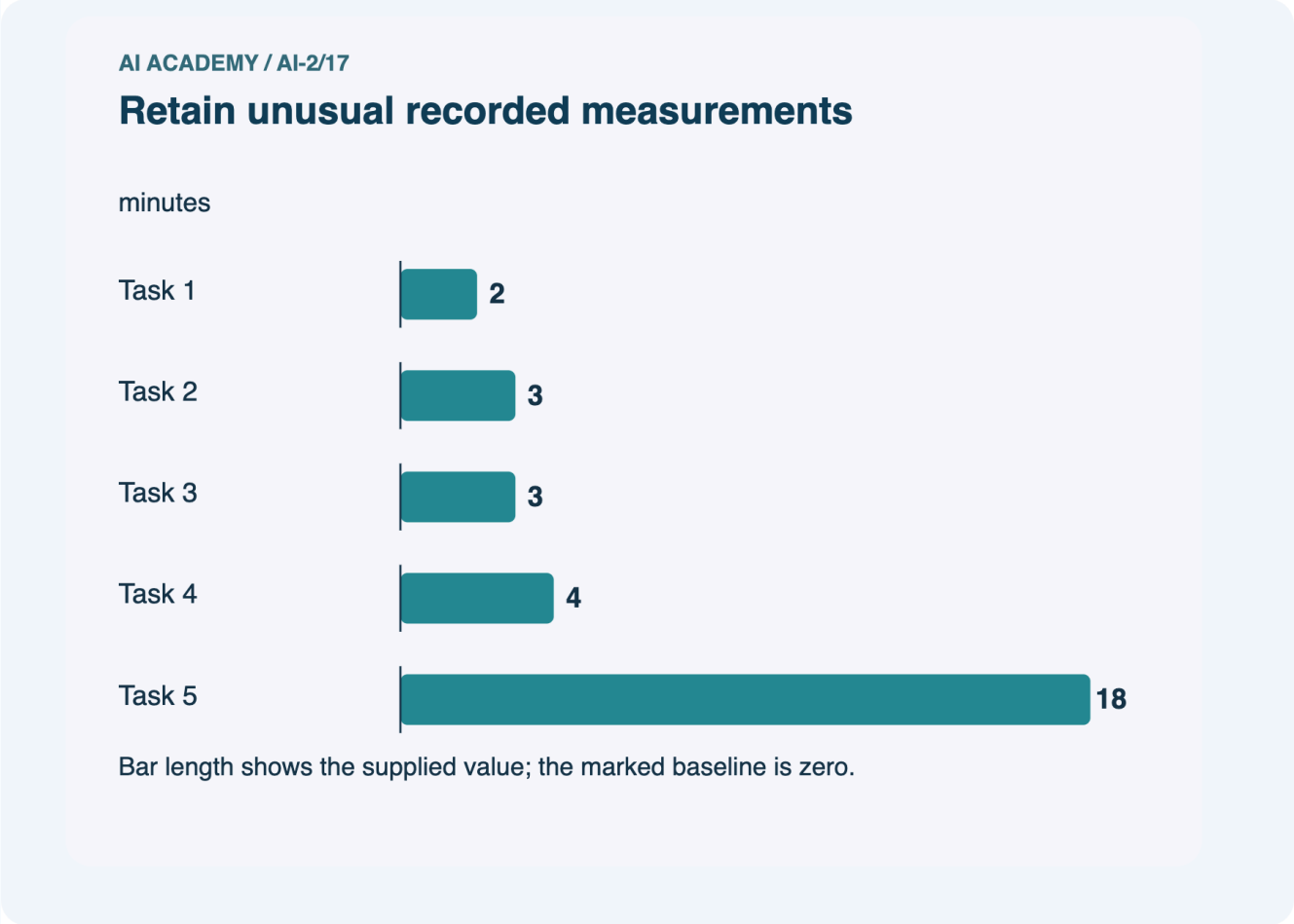
Follow the labelled picture with the matching explanation. Open the vector diagram

Mean: Sum 30; divide by 5: **mean** 6 minutes. → **Median**: The middle of five sorted values is 3 minutes. → **Investigate** 18: Check the source record and whether an unusually slow task occurred. → Document repair: Change a value only if **evidence** supports a correction; preserve the raw record.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Retain unusual recorded measurements



Follow the labelled picture with the matching explanation. Open the vector diagram

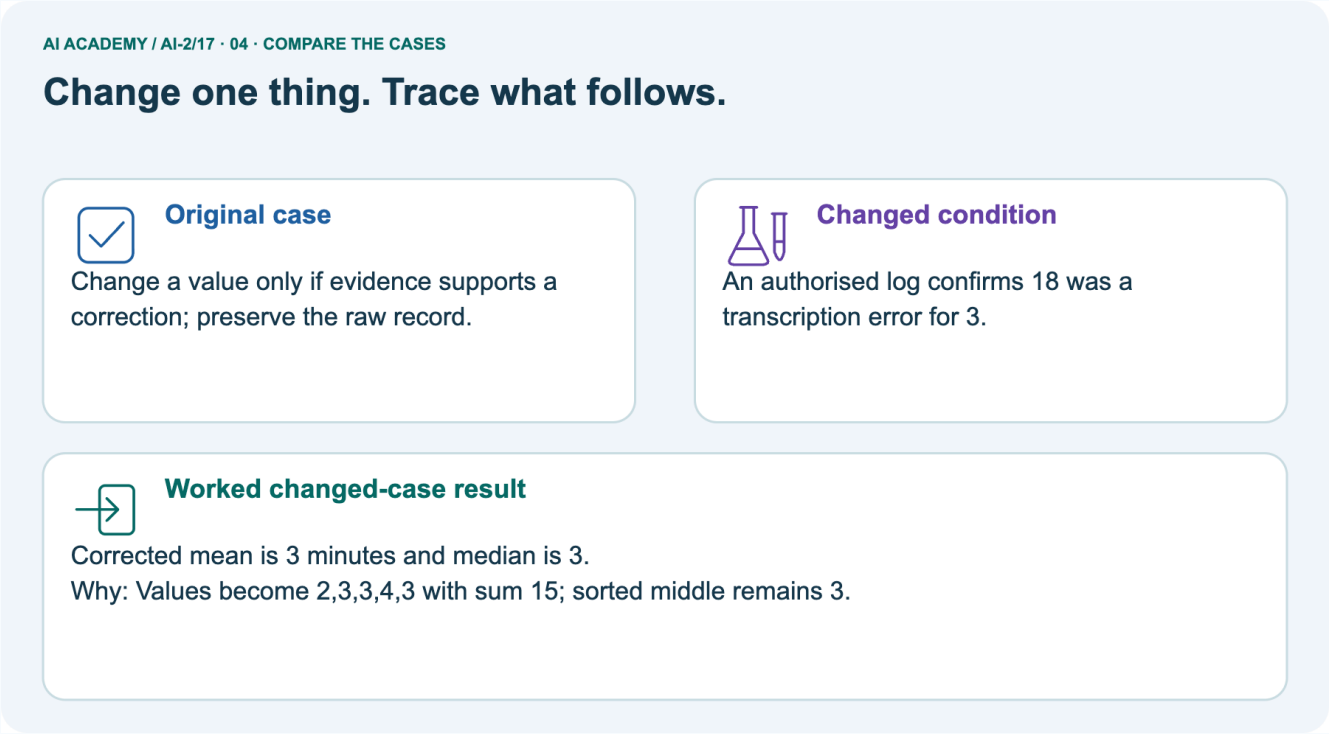
Retain unusual recorded measurements. Units: minutes. Task 1: 2. Task 2: 3. Task 3: 3. Task 4: 4. Task 5: 18.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column **mean**ings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Change a value only if **evidence** supports a correction; preserve the raw record. Changed condition: An authorised log confirms 18 was a transcription error for 3.

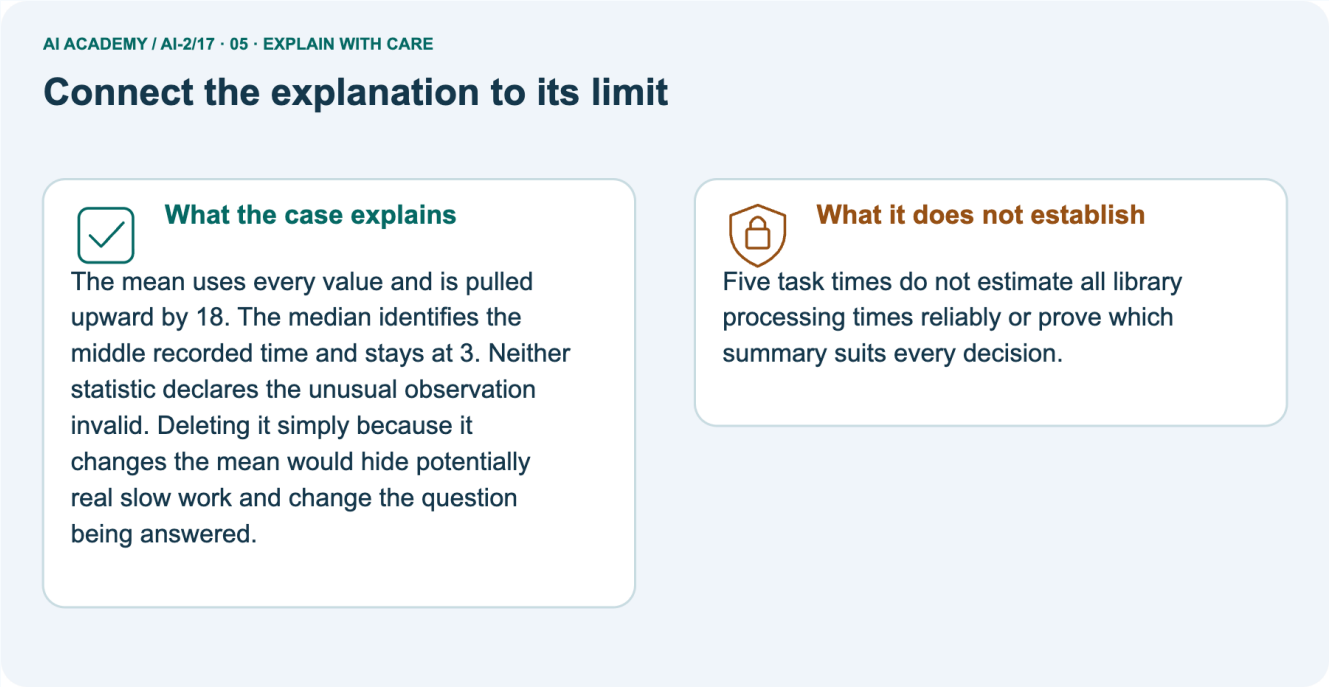
Expected result: Corrected **mean** is 3 minutes and **median** is 3. Reason: Values become 2,3,3,4,3 with sum 15; sorted middle remains 3.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Values become 2,3,3,4,3 with sum 15; sorted middle remains 3.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The **mean** uses every value and is pulled upward by 18. The **median** identifies the middle recorded time and stays at 3. Neither statistic declares the unusual **observation** invalid.

Deleting it simply because it changes the **mean** would hide potentially real slow work and change the question being answered.

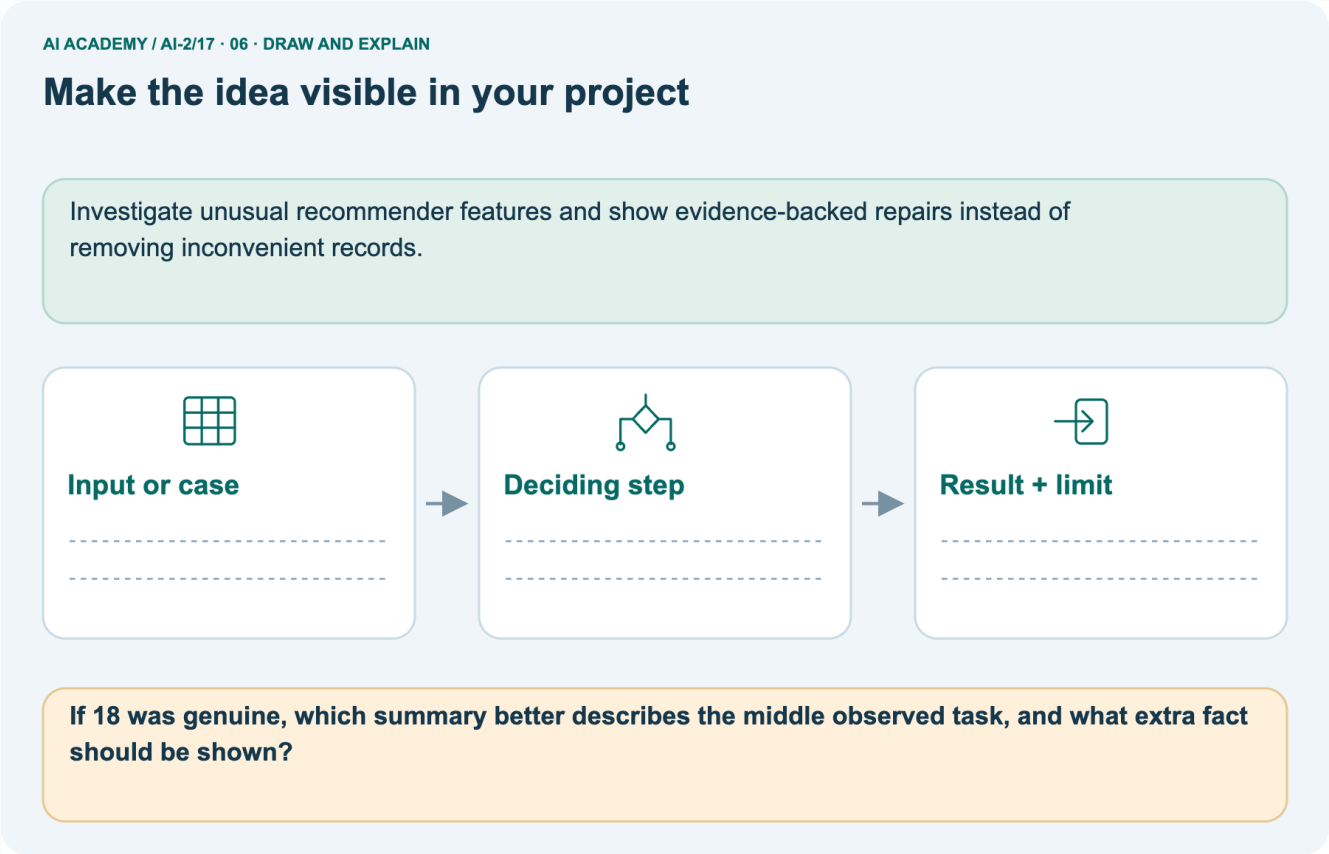
Limit: Five task times do not estimate all library **process**ing times reliably or prove which summary suits every decision.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Five task times do not estimate all library **process**ing times reliably or prove which summary suits every decision.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Investigate unusual recommender features and show **evidence**-backed repairs instead of removing inconvenient records. Draw your **input**, deciding step, result and limit.

Supported question: If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: **Median** 3 describes the middle; also report the unusual 18-minute task and the **mean** 6 rather than concealing spread.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Noise and Outliers: the case

02. Follow the deciding steps

03. Retain unusual recorded measurements

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.